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Multimodal ultra-high-field MRI, clinical, cognitive, and genetic profiles across the ALS–FTD spectrum

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Abstract

This dataset was acquired and curated to explore the spectrum of Motor Neuron Disease (MND) and Fronto-Temporal Dementia (FTD) with Ultra-High Field Magnetic Resonance Imaging (7 Tesla) and compare these to non-neurodegenerative disease (NND) controls (known colloquially as "The 7T hEalthy Ageing study [7TEA]"). Twenty people living with neurodegenerative disease and 14 NND controls underwent a comprehensive multimodal MRI protocol including structural, diffusion, quantitative MRI, resting state, and task fMRI, alongside cognitive testing and genetic

screening. This dataset combines detailed imaging phenotypes with extensive clinical characterisations. It facilitates investigations into the spectrum of MND and FTD, has provided a basis for developing novel quantitative biomarkers, and supports the exploration of interactions between imaging features and clinical progression. The availability of this dataset supports various research avenues, from detailed hippocampal subfield analyses, network connectivity assessments, and multimodal genetic, cognitive, and imaging studies. The dataset is published on OpenNeuro (dataset ds007036) and is curated in the Brain Imaging Data Structure (BIDS) standard.

Background & Summary

Motor Neuron Disease (MND) and Fronto-Temporal Dementia (FTD) are progressive neurodegenerative diseases, characterised by degeneration and death of neurons in often overlapping regions of the brain. FTD is an especially pernicious form of dementia due to its early onset, limited treatment options, and difficulty in diagnosis [1, 2]. Further, MND affects about 2,750 Australians in 2025, with numbers estimated to exceed 4,300 by 2050 [3]. In Australia, around two people diagnosed with MND die from the disease per day. Average survival is 27 months post-diagnosis, and the disease incurs enormous personal and societal cost, estimated at AUD 5.02 billion in 2025 [3].

People living with MND typically face a greater than 12-month delay between symptom onset and diagnosis [4]. A key reason for this delay is the lack of reliable biomarkers sensitive to disease onset and progression. Current diagnostic practices rely heavily on subjective clinical assessments, which lack the sensitivity and objectivity necessary for timely and accurate diagnosis and are insufficient for precise tracking of disease progression. MND biomarker development (and in particular, imaging biomarkers) in patients is essential due to the limited replication of disease characteristics in animal models [5].

While fronto-temporal lobar degeneration describes the broader underlying pathology that includes distinct protein-based neurodegenerative processes in the frontal and temporal lobes, FTD refers to the clinical syndrome, typically involving progressive behavioural, language, or executive dysfunction [6]. Similarly, MND is an umbrella term that includes distinct clinical phenotypes such as Amyotrophic Lateral Sclerosis (ALS; combined upper and lower motor neuron degeneration), primary lateral sclerosis (PLS; pure upper motor neuron involvement), and progressive muscular atrophy (PMA; pure lower motor neuron involvement), each with differing prognoses and disease progression. An Australian cohort study has demonstrated significant differences in survival and diagnostic delays across these subtypes [7].

MND and FTD overlap clinically, genetically, and pathologically, placing them along a spectrum rather than as entirely separate diseases [8]. Clinically, about 30–50% of ALS patients exhibit cognitive or behavioural deficits consistent with FTD, and around 10–15% meet full diagnostic criteria for FTD during their disease course. A subset (~15%) of individuals present with mixed features: motor neuron degeneration and frank FTD symptoms [9].

FTD typically manifests with early variable changes in behaviour, social cognition, language, or executive functioning. Presentations, especially the behavioural FTD variant, often mimic psychiatric disorders such as bipolar disorder, depression, or schizophrenia, especially in younger adults [1]. In contrast, MND's hallmark symptoms are characterised by progressive motor dysfunction: muscle weakness, spasticity, dysarthria, dysphagia, respiratory compromise, and fasciculations, reflecting upper and lower motor neuron loss [10]. Taken together, the overlap in symptoms and clinical presentations of MND and FTD present unique challenges for clinical diagnosis and prognosis.

Pathologically and genetically, both conditions share biological mechanisms. The most prevalent molecular hallmark is TDP-43 proteinopathy, accounting for ~97% of ALS and up to 45% of FTD cases [11]. A critical genetic link is the C9orf72 hexanucleotide repeat expansion, which is the most common inherited mutation found in both familial FTD and ALS, and it is reported to be responsible for approximately 25% of familial FTD, ~40% of familial MND, and 80% of ALS-FTD patients [8]. Notably, C9orf72 mutations can result in altered brain connectivity patterns, discernible on MRI [12].

Because presentations can evolve from pure motor to cognitive or vice versa and because early behavioural symptoms of FTD may precede motor signs of MND, early and precise differentiation of these diseases is essential for prognosis, clinical decision-making, and disease-specific interventions.

Advanced neuroimaging at ultra-high field (7 Tesla, 7T) MRI has emerged as a powerful tool capable of providing, high resolution, high signal-to-noise ratio (SNR) and detailed phenotyping, which is essential for individual-level diagnosis and longitudinal monitoring [13–16]. Both structural and functional changes have been reported in ALS-FTD, with UHF MRI emerging as a key tool for monitoring disease progression and biomarker development [15,17–23].

In response to challenges in developing specific disease biomarkers, we developed a study known colloquially as the 7T hEalthy Ageing (7TEA) study for earlier detection and differentiation of ALS-FTD through acquiring imaging, genetic, neuropsychological, and clinometric data. These data are vital for understanding the ALS-FTD spectrum and will assist in biomarker development and basic science applications.

Our dataset comprises 7T MRI data from 14 age-matched healthy NND controls and 20 people diagnosed across the ALS-FTD spectrum (**Fig. 1**), which includes:

I. Anatomical MRI: T1-weighted anatomical imaging using MP2RAGE for detailed structural analysis [24,25].
II. High-resolution hippocampal imaging: T2-weighted MRI for reliable hippocampal subfield segmentation [26,27].
III. Quantitative Susceptibility Mapping (QSM): Multi-Echo Gradient-Recalled Echo (ME-GRE) imaging (9 echoes) processed with QSMxT pipeline [28].
IV. Diffusion-weighted imaging (DWI): For white matter microstructural characterisation.
V. Functional MRI (fMRI): Resting-state fMRI and a naturalistic stimulus [29,30] paradigm utilising emotionally engaging movie clips from the silent film 'The Artist' [31] to potentially elicit brain responses relevant to FTD pathology (e.g., apathy).
VI. B1 mapping: For accurate T1 mapping, enhancing quantitative analyses [25].

In addition to imaging data, clinical and neuropsychological assessments and clinometric details were conducted, including:

I. The Addenbrooke's Cognitive Examination III (ACE-III) [32]

II. Measures of verbal and semantic fluency (FAS + animals) [33–36]
III. Hospital Anxiety and Depression Scale (HADS) [37]
IV. ALS Functional Rating Scale-Revised (ALSFRS-R) [38]
V. Formal diagnosis and clinical scores
VI. Genetic status (including C9orf72 expansions)

Derived and supporting imaging measures including:

I. Manual hypothalamus delineation (segmentation) [39]
II. PhysIO toolbox [40] regressors for physiological artifact correction in fMRI
III. Image quality assurance metrics derived from MRIQC [41].

This dataset combines detailed imaging phenotypes with extensive clinical characterisations. It facilitates investigations into structural and functional neural alterations associated with MND and FTD, provides a rich basis for developing novel quantitative biomarkers, and supports the exploration of complex interactions between imaging features and clinical progression. Furthermore, the integration of naturalistic fMRI paradigms offers new opportunities to study emotion processing and social cognition impairments characteristic of FTD [29].

The availability of this multimodal dataset facilitates several research directions, ranging from sophisticated network connectivity evaluations to in-depth hippocampus subfield analyses, or multimodal case-reports[42]. When combined with other publicly available datasets released from our site[43–45] and correctly harmonised[46] these data have great potential to assist in biomarker development for rare diseases such as MND. This dataset intends to improve early disease detection, improve diagnostic accuracy, inform disease-monitoring strategies, and contribute to future clinical management guidelines for people with MND and FTD by combining comprehensive imaging and clinical data.

Methods

Participants

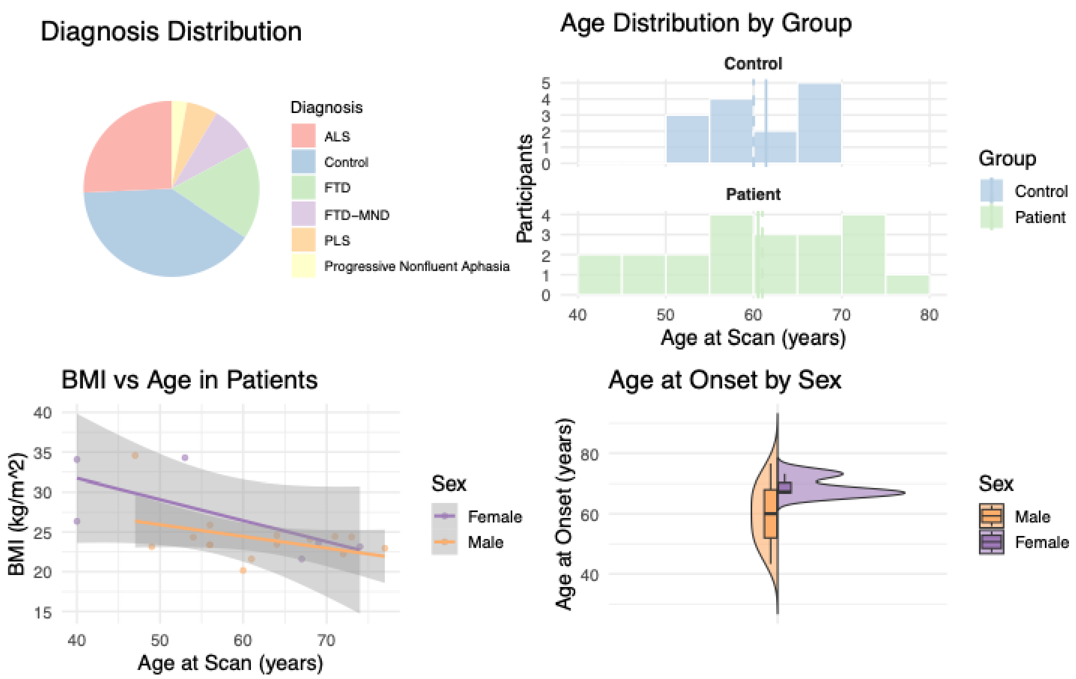
Inclusion criteria:

Participants were recruited from specialist neurological clinics across Queensland, Australia, and participants were included based on neuropsychological, electrophysiological, or imaging evidence of neurodegenerative diseases including MND and/or FTD. The present dataset includes participants with possible, probable, or definite MND as defined by the El Escorial criteria [47].

Participants initially included people with diagnoses of MND (n=13) [47] and FTD (n=7) [48,49], and 21 healthy age-matched NND controls. After updated clinical information was available, two participants had eventual diagnoses of primary lateral sclerosis (PLS) and Progressive Non-Fluent Aphasia. Three participants received a revised diagnosis of FTD-MND once updated clinical information was available to better inform a definitive diagnosis. This left ten patients with a final diagnosis of ALS, and six with a final diagnosis of FTD (**Figure 1**). Seven healthy control participants had either missing or incomplete data or did not consent to releasing their data publicly, leaving 14 remaining. The removal of 7 control participants limits the age and sex-matching of the dataset.

Demographics:
The mean age was 61.4 in NND controls and 60.8 in patients ($p = 0.828$). Average weight was 77.2kg in NND controls and 77.1kg in patients ($p = 0.975$) and average height was 169.2cm in NND controls and 175.2cm in patients ($p = 0.072$). The proportion of females to males was non-significant between NND controls and patients (χ^2 $p = 0.103$), though there were a larger proportion of females in the control group. Individual weight and height records are collapsed into 'body mass index' in the data records to mitigate risk of re-identification.

Figure 1: (Top left) distribution of diagnoses for participants included in this dataset; (top right) age distribution by diagnostic group (NND controls and patients); (bottom left) body mass index of patients split by sex; (bottom right) age at onset of symptoms stratified by sex for patients in this dataset.



Ethics statement

This study was approved by the relevant Human Research Ethics Committees including The Royal Brisbane and Women's Hospital (RBWH, HREC/EC00172) HREC and The University of Queensland HREC. Participants included in this dataset were provided written and informed consent to participate in the research and for deidentified data to be made available through publication, shared with other researchers, and for other research purposes. To protect the identities of the participants, pydeface (v2.02)[50] was used to de-identify all anatomical images. All identifiable metadata-based information was redacted. The dataset released with this work contains only de-identified data. The genetic data comprise a limited targeted mutation-status panel for ALS-associated variants, rather than genome-wide or sequence-level data. A contextual risk assessment of the combined dataset was performed prior to release, and accompanying demographic and clinical variables were minimised to reduce any residual risk of re-identification. Release of these data was undertaken with approval from the relevant data custodians.

Compliance with guidelines and regulations statement

All methods were performed in accordance with the National Statement on Ethical Conduct in Human Research (NHMRC, 2018), the Australian Code for the Responsible Conduct of Research (2018), and institutional policies of The University of Queensland, including the Responsible Research Management Framework and the Human Research Ethics Procedure. All investigators completed Good Clinical Practice (GCP) training prior to study commencement.

Imaging, Task and Assessments

During their research visit, participants completed two imaging sessions, each approximately 45 minutes in duration with a 15-minute break in between. Following arrival at the imaging facility, participants were made familiar with the MRI environment and the task.

In the first ~35-minute imaging session, structural image data were acquired viz: a 3-dimensional PETRA [51] sequence for coil combination, a $B1^+$ map for downstream accurate T1 map calculation [52], a T_1-weighted MP2RAGE sequence [24,25] for morphological analysis, a 3D Gradient-Recalled Echo (GRE) sequence for QSM estimation, and three repetitions of a Turbo-Spin-Echo (TSE) sequence for hippocampal segmentation [27].

After a short break, the second ~30-minute imaging session was conducted. Functional image data was acquired with a simultaneous multi-slice (SMS) echo planar imaging (EPI) sequence using two paradigms, resting state: where participants were instructed to keep their eyes open, and a viewing task: where participants watched the second and third acts of the 2011 silent film, *The Artist*. In preparation for the functional task, the participants watched an abridged version of the film's first act during the acquisition of the structural images. The movie viewing task was a condensed version of the finale of the film (See supplementary materials), focused on the emotional beats of the rising action, climax, and resolution of the naturalistic film and aimed to elicit a reliable emotional response from the participants [29,30]. Participants were also equipped with Electrocardiogram (ECG) electrodes and a breathing belt to record physiological activity. Finally, a 2D EPI diffusion-weighted imaging scan was acquired to probe white matter microstructural changes along the FTD/MND spectrum.

Image Acquisition Parameters

All scans were acquired on a 7-Tesla (7T) Siemens Magnetom research scanner (software baseline VB17A; Siemens Healthcare, Erlangen, Germany) at the Centre of Advanced Imaging, University of Queensland, using a 7T 1 channel transmit/32 channel (1Tx/32Rx) head array (Nova Medical, Wilmington, MA, USA). Although the sample size of this study is relatively modest, the higher Signal-to-Noise Ratio (SNR), contrast and resolution allow for a detailed picture of patient-specific pathologies, accommodating the variability in disease presentation in MND and FTD [13]. Indeed, recent studies have detailed the statistical power improvements from using 7T compared to standard 3T data [16,23].

Representative images of the scan protocol are displayed in Figure 2 and were acquired as follows:

Ultra-short TE images were acquired at 1 mm isotropic resolution using a prototype 3-dimensional PETRA sequence [51] (TR / TE / Flip Angle / FOV / Acquisition Time / Bandwidth = 1.99 ms / 0.07 ms / 2° / 288 x 288 x 288 mm / 1 m:56 s / 755 Hz/Px).

T1-weighted (T1w) structural scans were acquired using a 3-dimensional MP2RAGE prototype sequence [24,25] with 0.9 mm isotropic resolution (TR / TE / TIs / Flip Angles / FoV / Acquisition Time / Bandwidth = 4300 ms / 2.5 ms / 840 ms, 2370 ms / 5°, 6° / 240 × 225 x 230 mm / 6 m:54 s / 250 Hz/Px). Scans were acquired as a single slab with 256 sagittal slices, positioned in the centre of the brain, with the phase encoding direction anterior-to-posterior. Acceleration was performed using GRAPPA [53] (R = 3) and 6/8 partial Fourier in phase and slice encoding direction (Figure 2(A)).

$B1^+$ field mapping was performed using a vendor-supplied prototype 3D SA2RAGE sequence (Siemens WIP 654) [52]. Imaging parameters were as follows: TR / TE / TI_1 / TI_2 / FA_1 / FA_2 / FOV / Voxel size / Bandwidth / Acquisition time = 2400 ms / 0.95 ms / 106 ms / 1800 ms / 6° / 10° / 256 × 240 × 200 mm^3 / 4.0 × 4.0 × 5.0 mm^3 / 490 Hz px^{-1} / 1 min 14 s. A 3D slab acquisition (40 slices, 5 mm thickness, 0 % oversampling) was used with non-selective saturation recovery preparation. The sequence generated two inversion images for subsequent computation of quantitative $B1^+$ maps (Figure 2(B)).

Multi-echo GRE images were acquired at 0.75 mm isotropic resolution with echo times based on Robinson et al. (Figure 2(C)) [54] (TR / TEs / Flip Angle / FOV / Acquisition Time / Bandwidth / Flow Compensation / Readout Mode = 25 ms / (0.51, 7.14, 9.18, 11.22, 13.26, 15.30, 17.34, 19.38, 21.42)ms / 13° / 210 x 181 x 120 mm / 7 m:52 s / 1120 Hz/Px / First Echo / Monopolar). Scans were acquired as a single slab with 160 sagittal slices, positioned in the centre of the brain, with the phase encoding direction right-to-left. Acceleration was performed using GRAPPA [53] (R = 2).

Hippocampal imaging: Three repetitions of high-resolution T2-weighted images were acquired using a prototype 2-dimensional TSE sequence with a voxel size of 0.4 x 0.4 x 0.8 mm^3 (TR / TE / Flip Angle / FOV / Acquisition Time / Bandwidth = 10300 ms / 102 ms / 135° / 220 x 165 x 58mm / 4 m:19 s / 196 Hz/Px). Scans were acquired as a single, thin slab with 72 slices, aligned orthogonally to the hippocampus, with the phase encoding direction right-to-left. Acceleration was performed using GRAPPA [53] (R = 2), see Figure 2(E).

Functional images (Figure 2(F)) were acquired using a 2D SMS-EPI prototype (C2P) sequence with 2.0 mm isotropic resolution provided by the Center for Magnetic Resonance Research (University of Minnesota, Minneapolis, MN, USA) (TR / TE / FOV / Flip Angle / Bandwidth / = 1700 ms / 26.8 ms / 228 × 228 mm / 45° / 1994 Hz). Scans were acquired as an axial slab with 60 slices with a distance factor of 10 % aligned with the AC – PC line, with the phase encoding direction anterior-to-posterior. Acceleration was performed using GRAPPA [53,55,56] (R = 2) and 7/8 partial Fourier in the phase encoding. The multi-band factor was 2.

During the resting state scan, 300 volumes with a total acquisition time of 8 m:42 s were acquired. During the movie viewing task, 466 volumes with a total acquisition time of 13 m:24 s were acquired.

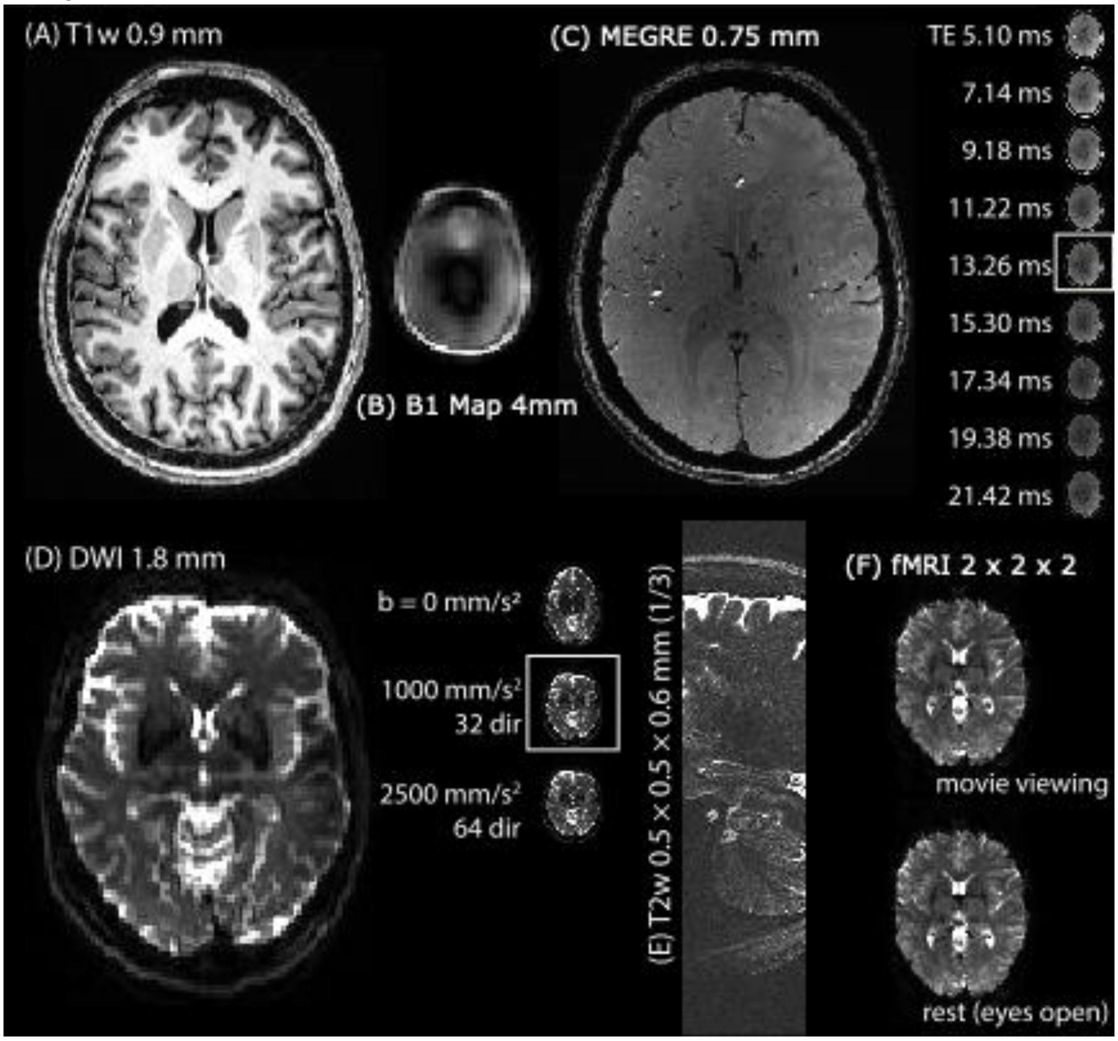
Diffusion weighted images (Figure 2 (D)) were acquired using a 2D EPI sequence with 1.8 mm isotropic resolution and b-values of 1000 and 2500 s/mm^2 and 30 and 60 directions, respectively (TR / TE / FOV / Flip Angle / Bandwidth / Readout Mode = 6400 ms / 64.4 ms / 216 × 216 mm / 180° / 1736 Hz / Monopolar). Scans were acquired as axially with 70 slices aligned with the AC – PC line, covering the cortex, with the phase encoding direction anterior-to-posterior. Acceleration was performed using GRAPPA [53] (R = 3) and 6/8 partial Fourier in the phase encoding. Acquisition time for b = 1000 s/mm^2 was 4 m:26 s, and 7 m:50 s for b = 2500 s/mm^2, respectively.

In addition, a total of 7 images with b = 0 s/mm^2 were also acquired including one with inverted phase encoding direction for distortion correction.

Quantitative Susceptibility Mapping (QSM) processing

The Multi-echo GRE was processed offline using the pipeline described in Bollmann et al. [57] to produce QSM image data. In brief, the 3D PETRA scan served as a short echo time reference scan for the combination of the individual coil images of the 3D Multi echo GRE data [58]. QSM values were estimated from the combined phase image using a total generalised variation (TGV) based QSM algorithm that incorporates phase unwrapping, background field removal, and dipole inversion in a single step [59].

Figure 2. Representative axial images from the scan protocol used in this dataset. (A) T1w MP2RAGE for anatomical delineation; (B) B1^{+} map for correction of the MP2RAGE T1 map; (C) Multi-echo gradient echo image with 9 echoes, including both magnitude and phase (not shown) for QSM processing; (D) Diffusion weighted imaging for white matter microstructural analysis; (E) sagittal slice of high resolution T2w Turbo Spin-Echo sequence for hippocampus subfield segmentation, three repetitions of a slab covering the hippocampus; (F) task-based (naturalistic movie viewing) and resting state fMRI acquisitions.



Physiological noise modelling

To account for physiological noise originating from respiration and cardiac activity, ECG and breathing belt data were acquired. An in-house algorithm was developed to remove gradient artefacts from the ECG signal. Then, the PhysIO toolbox was used to process ECG and pulse oximetry data [40], creating eight regressors for respiration and six regressors for cardiac activity.

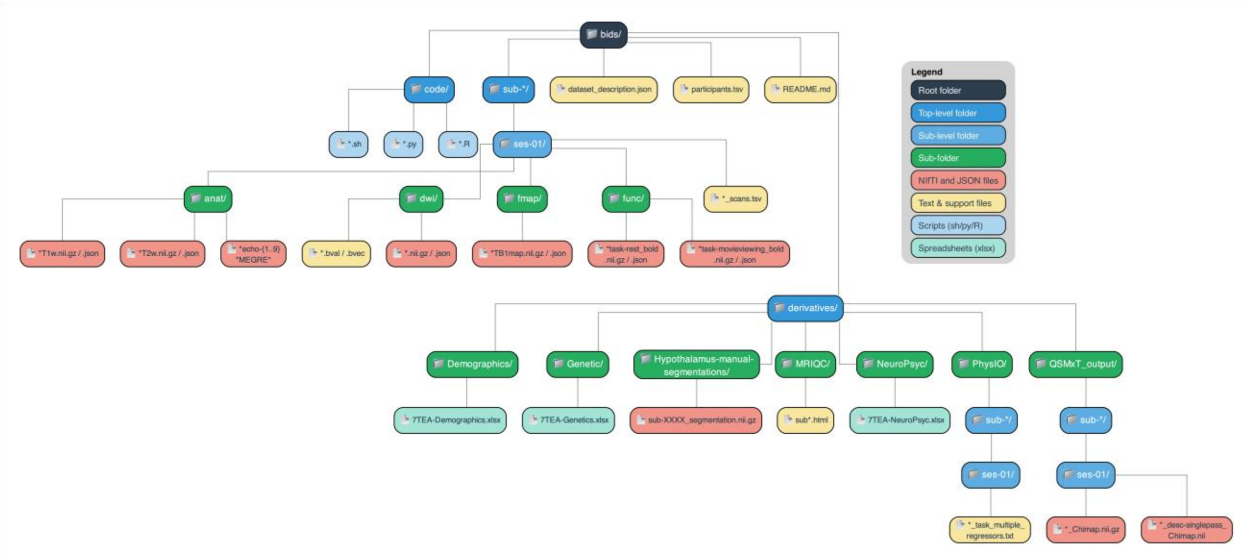
Data Records

This dataset (Accession number ds007036 [60]) has been deposited in OpenNeuro, an openly accessible neuroimaging repository containing numerous MRI datasets suitable for various research applications. In line with OpenNeuro requirements, the dataset adheres to the Brain

Imaging Data Structure (BIDS) v1.10.0 standard [61]. BIDS ensures organized, clear, and reproducible neuroimaging data, facilitating future research compatibility.

In compliance with BIDS, imaging data and corresponding JSON sidecar files are organized within modality-specific directories (e.g., anat, func), structured by sessions (e.g., ses-01) and subjects (e.g., sub-H0101, sub-P002). These sidecar files contain modality-specific metadata as detailed by the BIDS specification. Each directory, including recursive subdirectories, contains TSV files describing their contents, along with JSON files defining any non-standard terms. Phenotypic information and processed derivative data are housed under the derivatives directory. Any custom code used to generate or process the dataset is available in the code directory. All files follow the BIDS convention using standardized key-value pairs. An overview of the dataset organization is provided in Figure 3.

Figure 3: Dataset diagram of the published data and derivatives available on OpenNeuro (ID: ds007036). All data are organised as per the Brain Imaging Data Standard (BIDS) and includes derivatives of processed data.



Imaging Data

Under each subject directory are data from all MR contrasts including: T1w, T2w, DWI, fMRI, GRE, and B1 maps as described above. Four participants from the patient cohort have missing imaging data due to participant discomfort or protocol changes, including P0001 and P0002 (missing reverse phase-encoded DWI scan), P0017 (entire DWI missing due to error), P0008 could not complete any part of the scan due to discomfort.

MRIQC

MRI Quality Control (QC) of the data were conducted. The MRIQC (v24.0.0) [41] pipeline output is found under *derivatives*. The data has been de-identified. A summary of the processing is in the form of an HTML file in each subject directory, which contains values for quality control (see 'Technical Validation', below).

PhysIO

The physiological regressors generated by PhysIO are found under the *derivatives* folder. The final regressor matrix is saved as a text file that is compatible with SPM12 for first-level fMRI analysis.

Clinical measures

Our dataset includes demographics and clinical measures, stored under the *derivatives/demographics* folder. *Participant-level clinical data* included age, sex, body mass index at the time of scanning. Diagnostic information includes the final revised formal diagnosis (RDH). Disease severity for patients with ALS was measured using the revised ALS Functional Rating Scale (ALSFRS-R), a 48-point clinical tool quantifying functional impairment across bulbar, fine motor, gross motor, and respiratory domains [38]. Severity was binned using the standard formula of 48-score/time since onset (score is ALSFRS score and time since onset is the number of months): >1 is "fast", 0.5-1 is "medium" and <0.5 is "slow" progression. King's clinical staging was recorded, indicating the extent of anatomical spread of disease [62].

Genetic analysis

We targeted common pathogenic variants and risk alleles associated with MND. Genetic screening was performed by the Strategic ALS Australia Systems Genomics Consortium (SALSA-SGC) at The University of Queensland, which conducted targeted variant detection using established ALS gene panels.

Each column in the dataset corresponds to a specific gene and detection status. The screen included variants in SOD1, TARDBP, FUS, TBK1, and C9orf72, which represent the most frequently implicated causal genes in familial and sporadic ALS, as well as the UNC13A polymorphism, a well-established disease modifier [4,8,63]. Within SOD1 (superoxide dismutase 1), both pathogenic substitutions (e.g., G>T transitions) and known SNPs were assayed, each associated with toxic protein misfolding and motor neuron degeneration [64]. TARDBP (TAR DNA-binding protein 43) screening included variants known to disrupt RNA metabolism and protein homeostasis. FUS (fused in sarcoma) mutations were included due to their established role in early-onset and aggressive ALS phenotypes [65]. C9orf72 testing comprised the most common genetic cause of ALS and frontotemporal dementia[66]. UNC13A was included as a disease-modifying variant influencing survival and cognitive involvement[67]. TBK1 (TANK-binding kinase 1) mutation status was also recorded, with one participant carrying a truncating loss-of-function variant previously reported in this cohort [42].

Neuropsychological data

Data collected from clinical neuropsychologist (GR) are stored in the *derivatives/neuropsychological-data* directory. Three participants have missing data due to patient drop-out for their neuropsychological sessions. Two participants have incomplete data due to inability to complete the full session. Cognitive performance was assessed using the Addenbrooke's Cognitive Examination III (ACE-III), a 100-point screening tool evaluating attention, memory, verbal fluency, language, and visuospatial abilities. Corresponding domain scores and total scores are provided for each participant. Verbal fluency was further evaluated using both phonemic (F, A, S) and semantic (Animals) tasks [33–36], quantifying the number of unique words produced within one minute. Mood and affect were assessed using the Hospital Anxiety and Depression Scale (HADS) [37], which includes subscales for anxiety (HADS-A) and depression (HADS-D). Together, these measures provide a detailed neuropsychological profile for characterising cognitive and emotional changes across the ALS–FTD spectrum.

Manual hypothalamus segmentations

Hypothalami segmentation/delineations from a subset of the data were completed by author JC (see [39,68] for details). 10 segmentations for 5 randomly selected NND controls and 5 randomly selected people with MND can be found under *derivatives/hypothalamus-manual-segmentations* in nifti (.nii.gz) format. The hypothalamus is a critical brain structure important for regulating basic physiological functions such as appetite and sleep. Involvement of this structure in MND is increasingly recognised, modulating many non-motor symptoms of the disease; for review see Chang and colleagues (2015) [69]. Work examining the novel *TBK1* genetic mutation and lower hypothalamic volume [42], and work exploring the hypothalamus volumetry in MND [68] have been generated using these segmentations, with the full dataset and tool being developed in Ref. [39]. These segmentations include left and right hypothalamus and fornix.

Technical Validation

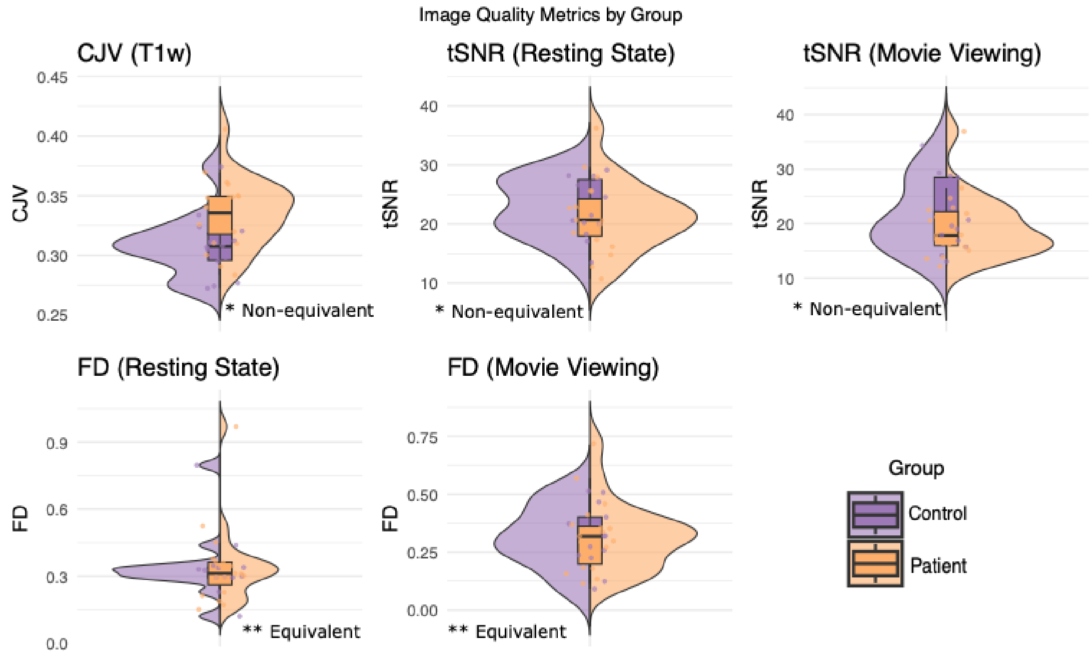
MRIQC

Summary statistics for the Image Quality Metrics (IQMs) are shown in Figure 4. The Coefficient of joint variation (CJV) is an objective function [70] for detecting head motion and artifacts. Lower values are indicative of higher image quality but are not expected to reach zero in real data. Temporal signal-to-noise ratio (tSNR) [71] indicates the sensitivity of the acquisition to identify small signal changes (higher is better), while Framewise Displacement (FD) shows instantaneous head motion [72] for both patients and NND controls.

To assess whether MRI quality differed meaningfully between patients and NND controls, we used the Two One-Sided Tests (TOST) procedure for equivalence testing, which evaluates whether group differences fall within a predefined negligible-effect range. Cohen's d bounds of

±0.8 were applied, the conventional threshold for a large effect size. Metrics included CJV, FD (rest/task), and tSNR (rest/task). See Figure 4 for each metric result. For each test, p_1 and p_2 represent the one-sided p-values from the TOST procedure, corresponding respectively to the lower and upper equivalence bounds; both must be below the significance threshold ($\alpha = 0.05$) to conclude that the observed group difference lies entirely within the predefined equivalence range (here, ±0.8 d). For FD during rest ($p_1 = 0.0174$; $p_2 = 0.0150$) and task ($p_1 = 0.0194$; $p_2 = 0.0138$) were within equivalence bounds, suggesting similar head motion across groups. tSNR (rest: $p_1 = 0.0033$, $p_2 = 0.0603$; task: $p_1 = 0.0023$, $p_2 = 0.111$) and CJV ($p_1 = 0.700$, $p_2 < 0.00001$) did not meet the equivalence threshold. Although these differences indicate somewhat reduced image quality in the patient group, this is typical of studies incorporating patients with symptoms affecting breathing, swallowing, and saliva production.

Figure 4. Image quality metric split-half violin plots by group (patient vs control). These metrics show the quality of scans, as described by MRIQC. (Top left) coefficient of joint variation (CJV); (top middle) temporal signal-to-noise ratio (tSNR) for resting state fMRI and (top right) naturalistic movie viewing fMRI; (bottom left) framewise displacement (FD) of resting state and (bottom middle) naturalistic fMRI tasks, showing instantaneous movement within the scan acquisition.



DWI quality metrics

Data were assessed for head movement during acquisition. First, skull-stripping was performed on each participant's T1-weighted image using FreeSurfer's SynthStrip[73]. The resulting brain mask was registered to the mean b=0 volume of the b=1000 AP series using FSL FLIRT[74]. The registered binary mask was used for all subsequent tSNR and motion computations. DWI data were corrected for eddy currents and head motion using FSL Eddy.[75] Outlier slices were detected using FSL Eddy's outlier replacement (--repol) with a linear second-level model (--slm=linear). tSNR was computed from the raw (pre-eddy-correction) b=0 volumes within each series. For each voxel, tSNR was defined as the ratio of the mean to the standard deviation across all b=0 volumes. Mean and median tSNR were summarised within the brain mask.

Data quality was high across both groups and both diffusion shells (Table 1) with expected reduction in signal-to-noise ratio (SNR) at higher b-values. Head motion was low in both groups. At b=1000, mean absolute displacement was 0.48 ± 0.27 mm in NND controls and 0.54 ± 0.39 mm in patients. At b=2500, mean absolute displacement was 0.79 ± 0.39 mm in NND controls and 0.87 ± 0.22 mm in patients. The proportion of outlier slices flagged was similarly low across groups (See Table 1), indicating that eddy current and motion artefacts were minimal. Patients showed slightly greater head motion and variability than NND controls across both shells.

TOST equivalence tests were performed between the 14 NND controls and 16 patients. The equivalence bound was again set at 0.8. When motion metrics were averaged across shells at the subject level, equivalence was not established for mean absolute motion (p_1 = 0.091, p_2 = 0.005). To probe further, we explored participants for which DWI quality metrics were low. One patient (sub-P0001) exhibited markedly low b=0 tSNR at b=2500 (tSNR = 11.9), which is approximately two standard deviations below the patient group mean. Three additional patients (sub-P0006, sub-P0007, sub-P0020) showed notably lower b=2500 tSNR relative to their b=1000 values (tSNR range: 20.5 - 20.8 at b=2500 vs. 62.1 - 125.6 at b=1000), which may reflect acquisition-related signal loss at the higher b-value.

Table 1: Diffusion quality metric details for non-neurodegenerative controls and people living with ALS (patients) including temporal SNR (tSNR), and relative and absolute displacement metrics for both b1000 and b2500 shells.

Shell	*Metric*	*NND Control (mean)*	*NND Control (SD)*	*Patient (mean)*	*Patient (SD)*
b=1000	b0 tSNR	87.959	17.519	84.89	24.98
b=1000	Mean absolute displacement (mm)	0.482	0.272	0.537	0.395
b=1000	Mean relative displacement (mm)	0.301	0.066	0.361	0.129
b=1000	Max absolute displacement (mm)	0.932	0.42	1.059	0.542
b=1000	Outlier slices (%)	1.15%	1.49%	1.61%	1.88%
b=2500	b0 tSNR	42.51	11.101	37.808	16.836
b=2500	Mean absolute displacement (mm)	0.787	0.386	0.872	0.221
b=2500	Mean relative displacement (mm)	0.467	0.055	0.563	0.19
b=2500	Max absolute displacement (mm)	1.534	0.669	1.636	0.283
b=2500	Outlier slices (%)	0.1	0.147	0.116	0.212

QSM quality metrics

Each participant's processed susceptibility map (Chimap) was evaluated within a brain mask derived from FreeSurfer SynthStrip (see above) applied to the Chimap. Tissue segmentation was performed on each participant's T1-weighted image using FSL FAST[76] with three tissue classes (CSF, grey matter, white matter). Partial volume estimates were thresholded at >0.9 to create tissue-specific binary masks. The following quality metrics were computed for each participant: (1) Whole-brain susceptibility statistics: mean ($\overline{x}$), standard deviation (σ), and interquartile range (IQR) of susceptibility values within the brain mask. (2) White matter (WM) to grey matter (GM) contrast-to-noise ratio (CNR), defined as:

$$\frac{\bar{x}_{GM} - \bar{x}_{WM}}{\sqrt{(0.5 * (\sigma_{GM}^2 + \sigma_{WM}^2)}}$$

which quantifies the ability of the susceptibility map to resolve the expected paramagnetic shift between grey and white matter. (3) Outlier voxel fraction: the percentage of brain voxels with absolute susceptibility exceeding 0.5 ppm, as an indicator of streaking artefacts or processing failures. See Table 2 for metric results.

To assess whether susceptibility map quality was equivalent between groups, TOST was employed as before. The WM-GM CNR was equivalent between groups (controls: 0.087 ± 0.048; patients: 0.080 ± 0.047; d = 0.15, TOST $p_1 = 0.008$, $p_2 = 0.045$), confirming that tissue contrast in the susceptibility maps was adequate for both groups. Overall, patients exhibited modestly greater susceptibility variance (potentially reflecting pathology-related susceptibility changes rather than data quality differences) than NND controls.

Table 2: Whole-brain QSM metrics for patients and NND controls including PPM values, outlier voxels, and grey and white matter (GM/WM) descriptive statistics for the cohort, plus contrast-to-noise ratio (CNR).

Metric	*NND controls (mean)*	*NND controls (SD)*	*Patient (mean)*	*Patients (SD)*
Whole brain mean (ppm)	0.0001	0.0001	0.0001	0.0001
Whole brain SD (ppm)	0.0227	0.0021	0.0261	0.0061
Whole brain IQR (ppm)	0.0191	0.0022	0.0229	0.007
Outlier voxels (%)	0.0014	0.0022	0.0026	0.0027
WM mean (ppm)	-0.0003	0.0007	0.0004	0.0013
GM mean (ppm)	0.0006	0.0017	-0.0008	0.0021
WM-GM CNR	0.0871	0.0477	0.08	0.0472

Code Availability

The code used to generate derivative datasets, figures, and statistics can be found under the *Code* folder of the repository on OpenNeuro. Code for generating QSMxT results can be found in the *QSMxT/code* folder. Various scripts in this directory were used in organising data and generating plots for this manuscript.

Data Availability Statement

All data have been deposited in OpenNeuro, an openly accessible neuroimaging repository and are available at doi:10.18112/openneuro.ds007036 [60]. All data are fully de-identified and MRI data are defaced.

Usage Notes

Additional genetic data details are available at https://salsasgc.org/collaborate/how-to-apply/.

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Competing interests

The authors have nothing to declare.

Author CRediT Statement

TBS: Methodology, Software, Validation, Formal analysis, Data Curation, Writing - Original Draft, Writing - Review & Editing, Project administration, Visualisation
AAN: Investigation, Methodology, Writing - Review & Editing
MB: Conceptualization, Methodology, Writing - Review & Editing, Resources, Supervision
StB: Formal analysis, Writing - Review & Editing

PB: Formal analysis, Writing - Review & Editing
JC: Data Curation, Formal analysis, Writing - Review & Editing
HDJ: Writing - Original Draft, Writing - Review & Editing
AF: Writing - Review & Editing
JF: Methodology, Software, Writing - Review & Editing, Project administration, Funding acquisition, Investigation, Conceptualisation, Supervision
NG: Conceptualization, Methodology, Writing - Review & Editing, Resources
CCG: Methodology, Project administration, Funding acquisition, Investigation, Conceptualisation, Supervision, Resources
RDH: Project administration, Funding acquisition, Investigation, Conceptualisation, Supervision, Project Administration, Writing - Review & Editing, Resources
EK: Data curation, Investigation
JL: Formal analysis, Data Curation, Methodology
PAM: Investigation, Conceptualisation, Supervision, Writing - Review & Editing, Resources
AN: Software, Resources, Data curation
STN: Resources, Data curation, Project administration, Writing - Review & Editing
VN: Methodology, Investigation, Project Administration
KO: Methodology, Software, Investigation, Resources, Project administration, Validation, Writing - Review & Editing
GR: Conceptualization, Methodology, Writing - Review & Editing
SR: Methodology, Software, Validation, Data curation, Writing - Review & Editing
OS: Project administration, Funding acquisition, Investigation, Conceptualisation, Supervision, Project Administration, Writing - Review & Editing, Resources
AS: Methodology, Software, Validation, Data curation, Writing - Review & Editing
FJS: Resources, Data curation, Project administration, Writing - Review & Editing
SaB: Methodology, Software, Validation, Formal analysis, Data Curation, Writing - Original Draft, Writing - Review & Editing, Project administration, Visualisation

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